2023-Paper

m-dependent Potential

$$V_m^-(x) = V(x) - 2\left[\frac{\mathcal{H}_m''}{\mathcal{H}_m} - \left(\frac{\mathcal{H}_m'}{\mathcal{H}_m}\right)^2 + 1\right], \tag{1}$$

$$\psi_{0,m}^-(x) = \left(\frac{2^m m!}{\sqrt{\pi}}\right)^{\frac{1}{2}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)}, \tag{2}$$

$$\psi_{n+1,m}^-(x) = \frac{1}{\sqrt{E_{n,m}^+ \; 2^n n! \sqrt{\pi}}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)} y_{n+1}^m(x) \;, \qquad n = 0, 1, 2..., \tag{3}$$

$$E_{n+1,m}^- = 2(n+m+1), \quad n = 0, 1, 2, \cdots, \quad \text{and } E_{0,m}^- = 0. \tag{4}$$

m and λ-dependent Potential

$$\hat{V}_m^-(\lambda, x) = V_m^-(x) - 2\frac{d^2}{dx^2} \ln\left[\mathcal{I}_m(x) + \lambda\right], \tag{5}$$

$$\mathcal{I}_m(x) = \left(\frac{2^m m!}{\sqrt{\pi}}\right) \int_{-\infty}^{x} \left[\frac{e^{-\frac{x'^2}{2}}}{\mathcal{H}_m(x')}\right]^2 dx'. \tag{6}$$

$$\hat{\psi}_{0,m}^-(\lambda, x) = \sqrt{\lambda(1+\lambda)}\, \frac{\psi_{0,m}^-(x)}{\mathcal{I}_m(x) + \lambda}\,. \tag{7}$$

$$\begin{aligned} \hat{\psi}_{n+1,m}^-(\lambda, x) &= \frac{1}{E_{n+1,m}^-} \hat{A}^\dagger A \psi_{n+1,m}^-(x), \qquad n = 0, 1, 2, \cdots, \\ &= \frac{e^{-\frac{x^2}{2}}}{\sqrt{2(n+m+1)2^n n! \sqrt{\pi}}} \left(\frac{y_{n+1}^m(x)}{\mathcal{H}_m(x)} + H_n(x) \frac{d}{dx} \ln\left[\mathcal{I}_m(x) + \lambda\right]\right) \end{aligned} \tag{8}$$

When $\lambda = 0$ or 1 (P or AM) the spectrum is strictily isospectral to the spectrum of starting potential.

$$E_{n,m}^+ = 2(n+m+1), \quad n = 0, 1, 2, \cdots \tag{9}$$

2024-Paper

$$V^+(x) = \frac{1}{4}\omega_x^2 x^2, \qquad -\infty < x < \infty, \tag{10}$$

$$\psi_n^+(x) \propto e^{-\frac{\omega_x}{4}x^2} H_n\left(\sqrt{\frac{\omega_x}{2}}x\right)$$

$$E_n^+ = \left(n + \frac{1}{2}\right)\omega_x\ , \quad n = 0, 1, 2, \cdots \tag{11}$$

$$V^-(x,m) = V^+(x) - 2\left[\frac{\mathcal{H}_m''\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} - \left[\frac{\mathcal{H}_m'\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}\right]^2 + \frac{\omega_x}{2}\right], \quad -\infty < x < \infty \tag{12}$$

$$\begin{aligned} \psi_0^-(x,m) &= \phi_m^{-1}(x) \propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}\ , \qquad m = 0, 2, 4, \cdots \\ \text{and} \quad \psi_{n+1}^-(x,m) &\propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} \hat{H}_{n+1,m}\left(\sqrt{\frac{\omega_x}{2}}x\right)\ , \qquad n = 0, 1, 2, \cdots \end{aligned} \tag{13}$$

$$\hat{H}_{n+1,m}(z) = \left[\mathcal{H}_m(z) H_{n+1}(z) + H_n(z)\frac{d}{dz}\mathcal{H}_m(z)\right], \tag{14}$$

$$E_{n+1,m}^- = E_{n,m}^+ = E_n^+ - \epsilon_m = (n+m+1)\omega_x \quad \text{with} \quad E_{0,m}^- = 0\,. \tag{15}$$

$$V_h^+(x) = \begin{cases} \frac{1}{4}\omega_x^2 x^2 & , \qquad x > 0 \\ \infty & , \qquad x \le 0\,. \end{cases} \tag{16}$$

$$\psi_{h,n}^+(x,\alpha) \propto x^{\alpha+\frac{1}{2}} e^{-\frac{\omega_x}{4}x^2} L_n^{(\alpha)}\left(\frac{\omega_x}{2}x^2\right); \quad n = 0, 1, 2, \cdots\,. \tag{17}$$

$$E_{h,n}^+(\alpha) = (2n+1+\alpha)\omega_x \qquad \text{and} \qquad \epsilon_{h,m}(\alpha) = -(2m+1+\alpha)\omega_x \tag{18}$$

$$E_{h,n,m}^+(\alpha) = E_{h,n}^+(\alpha) - \epsilon_{h,m}(\alpha) = 2(n+m+\alpha+1)\omega_x\,. \tag{19}$$

$$V_h^-(x,m,\frac{1}{2}) = V_h^+(x) - 2\left(\frac{\left[xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)\right]''}{xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)} - \left(\frac{\left[xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)\right]'}{xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)}\right)^2 + \frac{\omega_x}{2}\right); \qquad m = 0, 1, 2, \cdots\,. \tag{20}$$

$$V_h^-(x,m,-\frac{1}{2}) = V_h^+(x) - 2\left(\frac{L_m^{(-\frac{1}{2})''}\left(-\frac{\omega_x}{2}x^2\right)}{L_m^{(-\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)} - \left(\frac{L_m^{(-\frac{1}{2})'}\left(-\frac{\omega_x}{2}x^2\right)}{L_m^{(-\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)}\right)^2 + \frac{\omega_x}{2}\right); \qquad m = 0, 1, 2, \cdots\,. \tag{21}$$

$$\psi^-_{h,n}(x,m,\alpha) \propto x^{\alpha+\frac{3}{2}} e^{-\frac{\omega_x}{4}x^2} \frac{\hat{L}^{(\alpha)}_{n,m}(\frac{\omega_x}{2}x^2)}{L^{(\alpha)}_m\left(-\frac{\omega_x}{2}x^2\right)} \tag{22}$$

$$\hat{L}^{(\alpha)}_{n,m}(s) = L^{(\alpha)}_m(-s)L^{(\alpha+1)}_n(s) + L^{(\alpha+1)}_{m-1}(-s)L^{(\alpha)}_n(s)\,. \tag{23}$$

$$E^-_{h,n,m}(\alpha) = E^+_{h,n,m}(\alpha) = 2\,(n+m+1+\alpha)\,\omega_x\,. \tag{24}$$

In 2D

$$V^+(x,y) = \frac{1}{4}\omega_x^2 x^2 + \frac{1}{4}\omega_y^2 y^2; \quad -\infty < x < \infty, -\infty < y < \infty \tag{25}$$

$$\begin{aligned} \psi^+_{n_1,n_2}(x,y) &\propto \psi^+_{n_1}(x)\psi^+_{n_2}(y) \\ &\propto e^{-\frac{\omega_x x^2+\omega_y y^2}{4}} H_{n_1}\left(\sqrt{\frac{\omega_x}{2}}x\right) H_{n_2}\left(\sqrt{\frac{\omega_y}{2}}y\right) \qquad n_1, n_2 = 0,1,2,\cdots\,, \end{aligned} \tag{26}$$

$$E^+_{n_1,n_2} = E^+_{n_1} + E^+_{n_2} \tag{27}$$

$$V^-(x,y,m_1,m_2) = V^-(x,m_1) + V^-(y,m_2) \tag{28}$$

$$\begin{aligned} \psi^-_{0,0}(x,y,m_1,m_2) &= \phi^{-1}_{m_1}(x)\phi^{-1}_{m_2}(y), \\ \text{and} \quad \psi^-_{n_1+1,n_2+1}(x,y,m_1,m_2) &= \psi^-_{n_1+1}(x,m_1)\psi^-_{n_2+1}(y,m_2)\,, \qquad n_1,n_2 = 0,1,2,\cdots \end{aligned} \tag{29}$$

$$E^-_{n_1+1,n_2+1,m_1,m_2} = \left[(n_1+m_1+1)\,\omega_x + (n_2+m_2+1)\,\omega_y\right] \quad \text{and} \quad E^-_{0,0,m_1,m_2} = 0. \tag{30}$$

$$\psi^+_{h,n_1,n_2}(x,y,\alpha,\beta) \propto x^{\alpha+\frac{1}{2}} y^{\beta+\frac{1}{2}} e^{-\frac{\omega_x x^2+\omega_y y^2}{4}} L^{(\alpha)}_{n_1}\left(\frac{\omega_x}{2}x^2\right) L^{(\beta)}_{n_2}\left(\frac{\omega_y}{2}y^2\right); \quad n_1,n_2 = 0,1,2,\cdots \tag{31}$$

$$E^+_{h,n_1,n_2,\alpha,\beta} = \left[(2n_1+\alpha+1)\omega_x + (2n_2+\beta+1)\omega_y\right] - \epsilon_{h,m_1,m_2}, \tag{32}$$

$$V^-_h(x,y,m_1,m_2,\frac{1}{2}) = V^-_h(x,m_1,\frac{1}{2}) + V^-_h(y,m_2,\frac{1}{2}) \tag{33}$$

$$\psi^-_{h,n_1,n_2}(x,y,m_1,m_2,\frac{1}{2}) \propto \psi^-_{n_1}(x,m_1,\frac{1}{2})\psi^-_{n_2}(y,m_2,\frac{1}{2}) \tag{34}$$

$$\text{and} \quad E^-_{h,n_1,n_2}(m_1,m_2,\frac{1}{2}) = \left[(2n_1+2m_1+3)\,\omega_x + (2n_2+2m_2+3)\,\omega_y\right] \tag{35}$$

$$V_h^-(x,y,m_1,m_2,-\frac{1}{2}) = V_h^-(x,m_1,-\frac{1}{2}) + V_h^-(y,m_2,-\frac{1}{2}) \quad (36)$$

$$\psi_{h,n_1,n_2}^-(x,y,m_1,m_2,-\frac{1}{2}) \propto \psi_{h,n_1}^-(x,m_1,-\frac{1}{2})\psi_{h,n_2}^-(y,m_2,-\frac{1}{2}) \quad (37)$$

$$\text{and} \quad E_{h,n_1,n_2}^-(m_1,m_2,\frac{1}{2}) = [(2n_1+2m_1+1)\,\omega_x + (2n_2+2m_2+1)\,\omega_y]\ . \quad (38)$$

$$V_h^-(x,y,m_1,m_2,\frac{1}{2},-\frac{1}{2}) = V_h^-(x,m_1,\frac{1}{2}) + V_h^-(y,m_2,-\frac{1}{2}) \quad (39)$$

$$\psi_{h,n_1,n_2}^-(x,y,m_1,m_2,\frac{1}{2},-\frac{1}{2}) \propto \psi_{h,n_1}^-(x,m_1,\frac{1}{2})\psi_{h,n_2}^-(y,m_2,-\frac{1}{2}) \quad (40)$$

$$\text{and} \quad E_{h,n_1,n_2}^-(m_1,m_2,\frac{1}{2},-\frac{1}{2}) = [(2n_1+2m_1+3)\,\omega_x + (2n_2+2m_2+1)\,\omega_y] \quad (41)$$

$$V_{fh}^-(x,y,m_1,m_2,\beta) = V^-(x,m_1) + V_h^-(y,m_2,\beta), \quad (42)$$

$$\psi_{fh,0,0}^-(x,y,m_1,m_2,\beta) \propto \psi_0^-(x,m_1)\psi_{h,0}^-(y,m_2,\beta) \quad (43)$$

$$\psi_{fh,n_1+1,n_2}^-(x,y,m_1,m_2,\beta) \propto \psi_{n_1+1}^-(x,m_1)\psi_{h,n_2}^-(y,m_2,\beta); \quad (44)$$

$$E_{fh,0,0}^-(m_2,\beta) = 2(m_2+\beta+1)\omega_y. \quad (45)$$

$$\begin{aligned} E_{fh,n_1+1,n_2}^-(m_1,m_2,\beta) &= E_{n_1}^+(m_1) + E_{h,n_2}^+(m_2,\beta) \\ &= (n_1+m_1+1)\omega_x + 2(n_2+m_2+\beta+1)\omega_y \end{aligned} \quad (46)$$

$$V_{fh}^-(x,y,m_1,m_2,1/2) = V^-(x,m_1) + V_h^-(y,m_2,\frac{1}{2}) \quad (47)$$

$$\psi_{fh,0,0}^-(m_1,m_2,\frac{1}{2}) \propto \psi_0^-(x,m_1)\psi_{h,0}^-(y,m_2,\frac{1}{2}) \quad (48)$$

$$\psi_{n_1+1,n_2}^-(m_1,m_2,\frac{1}{2}) \propto \psi_{n_1+1}^-(x,m_1)\psi_{h,n_2}^-(y,m_2,\frac{1}{2}) \quad (49)$$

$$E_{0,0}^-(m_1,m_2,\frac{1}{2}) = (2m_2+3)\,\omega_y \quad (50)$$

$$E_{n_1+1,n_2}^-(m_1,m_2,\frac{1}{2}) = (n_1+2m_1+1)\,\omega_x + (2n_2+2m_2+3)\,\omega_y \quad (51)$$

$$V_{fh}^-(x,y,m_1,m_2,-1/2) = V^-(x,m_1) + V_h^-(y,m_2,-\frac{1}{2}) \quad (52)$$

$$\psi^{-}_{fh,0,0}(m_1,m_2,-\frac{1}{2}) \propto \psi^{-}_{0}(x,m_1)\psi^{-}_{h,0}(y,m_2,-\frac{1}{2}) \tag{53}$$

$$\psi^{-}_{n_1+1,n_2}(m_1,m_2,-\frac{1}{2}) \propto \psi^{-}_{n_1+1}(x,m_1)\psi^{-}_{h,n_2}(y,m_2,-\frac{1}{2}) \tag{54}$$

$$E^{-}_{0,0}(m_1,m_2,-\frac{1}{2}) = (2m_2+1)\,\omega_y \tag{55}$$

$$E^{-}_{n_1+1,n_2}(m_1,m_2,-\frac{1}{2}) = (n_1+2m_1+1)\,\omega_x + (2n_2+2m_2+1)\,\omega_y \tag{56}$$

In 3D

$$V^{+}(x,y,z) = \frac{1}{4}\omega^2(x^2+y^2) + \frac{1}{4}\omega_z^2 z^2; \quad -\infty < x,y,z < \infty \tag{57}$$

$$V^{+}(r,z) = \frac{1}{4}\omega^2 r^2 + \frac{1}{4}\omega_z^2 z^2 \tag{58}$$

$$\left[-\left(\frac{1}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\phi^2} + \frac{\partial^2}{\partial z^2}\right) + V^{+}(r,z)\right]\psi^{+}(r,\phi,z) = E\psi^{+}(r,\phi,z), \tag{59}$$

$$\psi^{+}(r,\phi,z) = \frac{e^{i\gamma\phi}\zeta(r,z)}{r^{\frac{1}{2}}}\,. \tag{60}$$

$$\zeta(r,z) = R(r)Z(z)\,. \tag{61}$$

$$\left[-\frac{\partial^2}{\partial r^2} - \frac{\partial^2}{\partial z^2} + V^{+}_{eff}(r,z)\right]\zeta(r,z) = E\,\zeta(r,z), \tag{62}$$

$$V^{+}_{eff}(r,z) = V^{+}(r,z) + \frac{\gamma^2 - 1/4}{r^2} \quad \text{with} \quad \gamma = 0,\pm1,\pm2,\cdots\,. \tag{63}$$

$$R(r) \to R_{n_1}(r) \propto r^{|\gamma|+\frac{1}{2}} e^{-\frac{\omega}{4}r^2} L^{(|\gamma|)}_{n_1}\left(\frac{1}{2}\omega r^2\right); n_1 = 0,1,2,\cdots \tag{64}$$

$$Z(z) \to Z_{n_2}(z) \propto e^{-\frac{\omega_z z^2}{4}} H_{n_2}\left(\sqrt{\frac{\omega_z}{2}}z\right); n_2 = 0,1,2,\cdots \tag{65}$$

$$E \to E_{n_1,n_2} = (2n_1+|\gamma|+1)\omega + \left(n_2+\frac{1}{2}\right)\omega_z\,. \tag{66}$$

$$E^{+}_{n_1,n_2} = E_{n_1,n_2} - \epsilon_{m_1,m_2} = 2(n_1+m_1+|\gamma|+1)\omega + (n_2+m_2+1)\omega_z. \tag{67}$$

$$
\begin{aligned}
V^{-}_{eff}(m_1,m_2,r,z) &= \frac{1}{4}\omega^2 r^2 + \frac{\gamma^2-\frac{1}{4}}{r^2} + \frac{2|\gamma|}{r^2} - (2m+1)\omega + \frac{1}{r^2} \\
&+ \frac{2r^2\omega^2 L^{(|\gamma|+1)}_{m_1-1}\left(-\frac{\omega r^2}{2}\right)^2}{L^{(|\gamma|)}_{m_1}\left(-\frac{\omega r^2}{2}\right)^2} \\
&+ \frac{\omega\left(\left(2|\gamma|+r^2\omega\right)L^{(|\gamma|+1)}_{m_1-1}\left(-\frac{\omega r^2}{2}\right) - r^2\omega L^{(|\gamma|+2)}_{m-2}\left(-\frac{\omega r^2}{2}\right)\right)}{L^{(|\gamma|)}_{m_1}\left(-\frac{\omega r^2}{2}\right)} \\
&+ \frac{1}{2}\omega_z^2 z^2 - 2\left[\frac{\mathcal{H}''_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} - \left[\frac{\mathcal{H}'_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}\right]^2 + \frac{\omega_z}{2}\right] \\
\zeta^{-}_{n_1,n_2+1}(r,z,m_1,m_2) &\propto r^{|\gamma|+\frac{3}{2}} e^{-\frac{\omega r^2}{4}} \frac{\hat{L}^{(\gamma)}_{n_1,m_1}\left(\frac{1}{2}\omega r^2\right)}{L^{(\gamma)}_{m_1}\left(-\frac{1}{2}\omega r^2\right)}\, e^{-\frac{\frac{\omega_z}{2}z^2}{2}} \frac{\hat{H}_{n_z+1,m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} \\
E^{-}_{n_1,n_2+1}(m_1,m_2) &= 2(n_1+m_1+|\gamma|+1)\omega + (n_2+m_2+1)\,\omega_z
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
\zeta^{-}_{0,0}(r,z,m_1,m_2) \quad &\propto \quad r^{|\gamma|+\frac{3}{2}} \frac{e^{-\frac{\omega r^2}{4}} L^{(|\gamma|+1)}_{\mathrm{m1}}\left(-\frac{1}{2}\omega r^2\right)\big)}{L^{(|\gamma|)}_{m_1}\left(-\frac{1}{2}\omega r^2\right)} \times \frac{e^{-\frac{\omega_z\,z^2}{4}}}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} \\
\text{and} \quad E^{-}_{0,0,m_1,m_2} \quad &= \quad 2(m_1+|\gamma|+1)\omega
\end{aligned} \tag{69}
$$

2025-Paper

$$V^l(x) = V(x) + \lambda_0 x, \qquad V(x) = \frac{1}{4}\omega_1^2 \tag{70}$$

$$\tilde{V}^l(\tilde{x}) = \frac{1}{4}\omega_1^2\tilde{x}^2 - \frac{\lambda_0^2}{\omega_1^2} \quad \text{where} \quad \tilde{x} = x + \frac{2\lambda_0}{\omega_1^2}, \tag{71}$$

$$E \to \tilde{E}_n^l(\omega_1) = \left(n + \frac{1}{2}\right)\omega_1 + \frac{\gamma_0^2}{\omega_1^2}; \quad n = 0, 1, 2... \tag{72}$$

$$\tilde{\psi}_n^l(\tilde{x}) = \psi_n(\tilde{x}), \quad \text{Here,} \quad \tilde{\omega_1} = \omega_1. \tag{73}$$

1D

$$V_{RE,m}(x) = V(x) + V_{rat,m}\left(x\right), \quad -\infty < x < \infty \tag{74}$$

$$V_{rat,m}(x) = -2\left[\frac{\mathcal{H}_m''\left(\sqrt{\frac{\omega_1}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} - \left[\frac{\mathcal{H}_m'\left(\sqrt{\frac{\omega_1}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)}\right]^2 + \frac{\omega_1}{2}\right] \tag{75}$$

$$\begin{aligned} \psi_{RE,m,0}(x) &= \frac{\zeta\left(\omega_1, x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} \\ \text{and} \quad \psi_{RE,m,n+1}(x) &= \frac{\zeta\left(\omega_1, x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)}\hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}x\right), \end{aligned} \tag{76}$$

$$\hat{H}_{m,n+1}(x) = \mathcal{H}_m(x)H_{n+1}(x) + H_n(x)\frac{d}{dx}\mathcal{H}_m(x), \tag{77}$$

$$E_{RE,m,n+1}(\omega_1) = \left(n + m + 1\right)\omega_1 \qquad \text{with} \quad E_{RE,m,0}(\omega_1) = 0. \tag{78}$$

$$\tilde{V}_{RE,m}^l(\tilde{x}) = \tilde{V}^l(\tilde{x}) + V_{rat,m}\left(\tilde{x}\right), \quad -\infty < x < \infty \tag{79}$$

$$\begin{aligned} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \frac{\zeta\left(\omega_1, \tilde{x}\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)} \\ \text{and} \quad \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= \frac{\zeta\left(\omega_1, \tilde{x}\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)}\hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right). \end{aligned} \tag{80}$$

$$\tilde{E}_{RE,m,n+1}^l(\omega_1) = \left(n + m + 1\right)\omega_1 \qquad \text{with} \quad \tilde{E}_{RE,m,0}^l(\omega_1) = 0. \tag{81}$$

2D

$$V(x,y) = \frac{1}{4}\omega_1^2 x^2 + \frac{1}{4}\omega_2^2 y^2 + \frac{\lambda}{2}xy. \tag{82}$$

$$\begin{matrix} x = a\tilde{x} + b\tilde{y}, \\ y = -b\tilde{x} + a\tilde{y}, \end{matrix} \begin{cases} \tilde{x} = ax - by, \\ \tilde{y} = bx + ay, \end{cases} \tag{83}$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\frac{1}{2}\left(\omega_1^2 + \omega_2^2 - \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2}\right)} \\ \text{and} \quad \tilde{\omega}_2 &= \sqrt{\frac{1}{2}\left(\omega_1^2 + \omega_2^2 + \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2}\right)} \end{aligned} \tag{84}$$

$$\tilde{V}(\tilde{x},\tilde{y}) = \tilde{V}(\tilde{x}) + \tilde{V}(\tilde{y}). \tag{85}$$

$$\tilde{\psi}_{n_1,n_2}(\tilde{x},\tilde{y}) \propto \tilde{\psi}_{n_1}(\tilde{x})\ \tilde{\psi}_{n_2}(\tilde{y}), \quad n_1, n_2 = 0, 1, 2, \cdots, \tag{86}$$

$$\text{and} \quad \tilde{E}_{n_1,n_2}(\tilde{\omega}_1,\tilde{\omega}_2) = \left(n_1 + \frac{1}{2}\right)\tilde{\omega}_1 + \left(n_2 + \frac{1}{2}\right)\tilde{\omega}_2, \tag{87}$$

$$\tilde{V}_{RE,m_1,m_2}(\tilde{x},\tilde{y}) = \tilde{V}_{RE,m_1}(\tilde{x}) + \tilde{V}_{RE,m_2}(\tilde{y}), \tag{88}$$

$$\tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x},\tilde{y}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x})\ \tilde{\psi}_{RE,m_2,0}(\tilde{y}) \tag{89}$$

$$\tilde{\psi}_{RE,m_1,m_2,n_1+1,n_2+1}(x,y) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x})\ \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y}) \tag{90}$$

$$\text{and} \quad \tilde{E}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{\omega}_1,\tilde{\omega}_2) = \tilde{E}_{RE,m_1,n_1+1}(\tilde{\omega}_1) + \tilde{E}_{RE,m_2,n_2+1}(\tilde{\omega}_2) \tag{91}$$

$$\text{with} \quad \tilde{E}_{RE,m_1,m_2,0,0}(\tilde{\omega}_1,\tilde{\omega}_2) = 0, \quad n_1, n_2 = 0, 1, 2......., $$

3D-Quadratic+Linear

$$V^{lq}(x,y,z) = \frac{1}{4}\left(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2\right) + \lambda_0 z + \frac{\lambda}{2}xy. \tag{92}$$

$$V^{lq}(x,y,z) = V(x,y) + V^l(z), \tag{93}$$

$$\tilde{V}^{lq}(\tilde{x},\tilde{y},\tilde{z}) = \tilde{V}(\tilde{x},\tilde{y}) + \tilde{V}^l(\tilde{z}). \tag{94}$$

$$\tilde{\psi}^{lq}_{n_1,n_2,n_3}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{n_1,n_2}(\tilde{x},\tilde{y})\tilde{\psi}^l_{n_3}(\tilde{z}) \tag{95}$$

$$\tilde{E}^{lq}_{n_1,n_2,n_3}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{n_1,n_2}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}_{n_3}(\tilde{\omega}_3) \tag{96}$$

$$\tilde{V}^{lq}_{RE,m_1,m_2,m_3}(\tilde{x},\tilde{y},\tilde{z}) = \tilde{V}_{RE,m_1,m_2}(\tilde{x},\tilde{y}) + \tilde{V}^{l}_{RE,m_3}(\tilde{z}) \tag{97}$$

$$\tilde{\psi}^{lq}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x},\tilde{y})\ \tilde{\psi}^{l}_{RE,m_3,0}(\tilde{z}), \tag{98}$$

$$\tilde{\psi}^{lq}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{x},\tilde{y})\ \tilde{\psi}^{l}_{RE,m_3,n_3+1}(\tilde{z}) \tag{99}$$

$$\tilde{E}^{lq}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{RE,m_1,m_2,n_1+1,n_2+1}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}^{l}_{RE,m_3,n_3+1}(\tilde{\omega}_3) \tag{100}$$

$$\tilde{E}^{lq}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = \tilde{E}_{RE,m_1,m_2,0,0}(\tilde{\omega}_1,\tilde{\omega}_2) + \tilde{E}^{l}_{RE,m_3,0}(\tilde{\omega}_3)$$

3D-Quadratic Only

$$V^q(x,z,y) = \frac{1}{4}\left(\ \omega^2(x^2+y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx\right). \tag{101}$$

$$V^{q_1}(x,y,z) = \frac{1}{4}\left(\omega^2(x^2+y^2) + \omega_z^2 z^2 + 2\lambda_2 yz + 2\lambda_3 zx\right) \tag{102}$$

$$\begin{aligned} x &= ad\tilde{y} + bd\tilde{z} - c\tilde{x}, \\ y &= d\,\tilde{x} + ac\tilde{y} + bc\tilde{z} \\ z &= a\tilde{z} - b\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = -cx + d\,y, \\ \tilde{y} = ad\,x + acy - bz, \\ \tilde{z} = bd\,x + bcy + az. \end{cases} \tag{103}$$

$$\tilde{V}^q(\tilde{x},\tilde{y},\tilde{z}) = \frac{1}{4}\left(\tilde{\omega}_1^2\tilde{x}^2 + \tilde{\omega}_2^2\tilde{y}^2 + \tilde{\omega}_3^2\tilde{z}^2\right) \tag{104}$$

$$\begin{aligned} \tilde{\omega}_1 &= \omega, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 - \sqrt{4(\lambda_2^2+\lambda_3^2) + (\omega^2-\omega_3^2)^2}\right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 + \sqrt{4(\lambda_2^2+\lambda_3^2) + (\omega^2-\omega_3^2)^2}\right)}. \end{aligned} \tag{105}$$

$$\tilde{V}^{q_1}_{RE,m_1,m_2,m_3}(\tilde{x},\tilde{y},\tilde{z}) = \tilde{V}_{RE,m_1}(\tilde{x}) + \tilde{V}_{RE,m_2}(\tilde{y}) + \tilde{V}_{RE,m_3}(\tilde{z}) \tag{106}$$

$$\tilde{\psi}^{q_1}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x})\ \tilde{\psi}_{RE,m_2,0}(\tilde{y})\ \tilde{\psi}_{RE,m_3,0}(\tilde{z})$$

$$\tilde{\psi}^{q_1}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{x},\tilde{y},\tilde{z}) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x})\ \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y})\ \tilde{\psi}_{RE,m_3,n_3+1}(\tilde{z}) \tag{107}$$

$$\begin{aligned}\tilde{E}^{q_1}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) &= \tilde{E}_{RE,m_1,n_1+1}\left(\tilde{\omega}_1\right) \\ &+ \tilde{E}_{RE,m_2,n_2+1}\left(\tilde{\omega}_2\right) + \tilde{E}_{RE,m_3,n_3+1}\left(\tilde{\omega}_3\right)\end{aligned} \tag{108}$$

$$\tilde{E}^{q_1}_{RE,m_1,m_2,m_3,0,0,0}(\tilde{\omega}_1,\tilde{\omega}_2,\tilde{\omega}_3) = 0 \tag{109}$$

$$V^{q_2}(x,y,z) = \frac{1}{4}\left(\omega^2(x^2+y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda(yz+zx)\right). \tag{110}$$

$$\begin{aligned} x &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} - \frac{\tilde{x}}{\sqrt{2}}, \\ y &= \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} + \frac{\tilde{x}}{\sqrt{2}}, \\ z &= a\tilde{z} - b\tilde{y}, \end{aligned} \begin{cases} \tilde{x} = \dfrac{y}{\sqrt{2}} - \dfrac{x}{\sqrt{2}}, \\ \tilde{y} = \dfrac{a}{\sqrt{2}}x + \dfrac{a}{\sqrt{2}}y - bz, \\ \tilde{z} = \dfrac{b}{\sqrt{2}}x + \dfrac{b}{\sqrt{2}}y + az. \end{cases} \tag{111}$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\omega^2 - \lambda_1}, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 + \lambda_1 - \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2}\right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2}\left(\omega^2 + \omega_3^2 + \lambda_1 + \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2}\right)}. \end{aligned} \tag{112}$$

Uncertainty Calculations: 2023-Paper

$$V_m^-(x) = V(x) - 2\left[\frac{\mathcal{H}_m''}{\mathcal{H}_m} - \left(\frac{\mathcal{H}_m'}{\mathcal{H}_m}\right)^2 + 1\right], \tag{113}$$

m	$n=-1$	$n=0$	$n=1$
0	0.5	1.5	2.5
2	≈ 0.5172	≈ 1.554	≈ 2.3412
4	≈ 0.5212	≈ 1.6152	≈ 2.2102

Table 1: Uncertainty relation $\Delta x\,\Delta p$ for the potential $V_m^-(x)$ in the ground state, first excited state, and second excited state for $m = 0, 2, 4$.

$$\hat{V}_m^-(\lambda, x) = V_m^-(x) - 2\frac{d^2}{dx^2}\ln\left[\mathcal{I}_m(x) + \lambda\right], \tag{114}$$

$m/\lambda =$	1×10^{-12}	1×10^{-8}	1×10^{-5}	1×10^{-3}	1×10^{-1}	1×10^{2}
0	≈ 0.5202	≈ 0.5281	≈ 0.5367	≈ 0.5590	≈ 0.5455	≈ 0.5000
2	≈ 0.5223	≈ 0.5270	≈ 0.5285	≈ 0.5266	≈ 0.5193	≈ 0.5172
4	≈ 0.5228	≈ 0.5252	≈ 0.5246	≈ 0.5232	≈ 0.5215	≈ 0.5212

Table 2: Uncertainty relation $\Delta x\,\Delta p$ for potential $\hat{V}_m^-(\lambda, x)$ in the ground state for $m = 0, 2, 4$.

Uncertainty Calculations: 2024-Paper

Full Line Oscillator

$$V^-(x,m) = V^+(x) - 2\left[\frac{\mathcal{H}''_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} - \left[\frac{\mathcal{H}'_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}\right]^2 + \frac{\omega_x}{2}\right], \quad -\infty < x < \infty$$

$n\backslash m$	0	2	4	6
0	0.5	0.517247	0.521212	0.522388
1	1.5	1.55415	1.61522	1.66868
2	2.5	2.34124	2.21019	2.12295
3	3.5	3.50248	3.53107	3.56781

Table 3: Uncertainty values for Full Line Oscillator.

Half Line Oscillator

$$V_h^-(x,m,-\frac{1}{2}) = V_h^+(x) - 2\left(\frac{L_m^{(-\frac{1}{2})''}\left(-\frac{\omega_x}{2}x^2\right)}{L_m^{(-\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)} - \left(\frac{L_m^{(-\frac{1}{2})'}\left(-\frac{\omega_x}{2}x^2\right)}{L_m^{(-\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)}\right)^2 + \frac{\omega_x}{2}\right); \qquad m = 0, 1, 2, \cdots.$$

$n\backslash m$	0	1	2	3
0	0.583216	0.68377	0.747385	0.796063
1	1.49105	1.43835	1.44864	1.46897
2	2.37291	2.29944	2.28775	2.29239
3	3.24886	3.17543	3.15381	3.14931

Table 4: Uncertainty values for Half Line Oscillator with $\alpha = -\frac{1}{2}$.

$$V_h^-(x,m,\frac{1}{2}) = V_h^+(x) - 2\left(\frac{\left[xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)\right]''}{xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)} - \left(\frac{\left[xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)\right]'}{xL_m^{(\frac{1}{2})}\left(-\frac{\omega_x}{2}x^2\right)}\right)^2 + \frac{\omega_x}{2}\right); \qquad m = 0, 1, 2, \cdots.$$

$n\backslash m$	0	1	2	3
0	0.525237	0.53416	0.539833	0.543865
1	1.46161	1.45048	1.44748	1.44709
2	2.36972	2.34908	2.33979	2.33524
3	3.26505	3.24028	3.22724	3.21972

Table 5: Uncertainty values for Half Line Oscillator with $\alpha = \frac{1}{2}$

$\mathcal{I}_m$ for Full and Half Line Oscillator

m	$\mathcal{I}_m(x)$
0	$\frac{1}{2}(\mathrm{erf}(x)+1)+\lambda$
2	$\frac{1}{2}\left(\mathrm{erf}(x)+\frac{2e^{-x^2}\,x}{\sqrt{\pi}(2x^2+1)}+1\right)+\lambda$
4	$\frac{1}{2}\left(1+\mathrm{erf}(x)+\frac{2e^{-x^2}\,x(5+2x^2)}{\sqrt{\pi}(3+4x^2(3+x^2))}\right)+\lambda$

Table 6: Expressions for $\mathcal{I}_m(x)$ in the Full line Oscillator for $m = 0, 2, 4$ with $\omega_x = 2$.

m	$I_m(x)$
0	$-\frac{2e^{-x^2}x(3+2x^2)}{3\sqrt{\pi}}+\mathrm{erf}(x)+\lambda$
1	$-\frac{2e^{-x^2}x(15+20x^2+4x^4)}{5\sqrt{\pi}(3+2x^2)}+\mathrm{erf}(x)+\lambda$
2	$-\frac{2e^{-x^2}x(105+210x^2+84x^4+8x^6)}{7\sqrt{\pi}(15+20x^2+4x^4)}+\mathrm{erf}(x)+\lambda$
3	$-\frac{2e^{-x^2}x(945+2520x^2+1512x^4+288x^6+16x^8)}{9\sqrt{\pi}(105+210x^2+84x^4+8x^6)}+\mathrm{erf}(x)+\lambda$

Table 7: Expressions for $I_m(x)$ in the Half Line Oscillator with $\alpha = \frac{1}{2}$, for $m = 0, 1, 2, 3$ with $\omega_x = 2$.

m	$I_m(x)$
0	$-\frac{2e^{-x^2}x}{\sqrt{\pi}}+\mathrm{erf}(x)+\lambda$
1	$-\frac{2e^{-x^2}x\,(3+2x^2)}{3\sqrt{\pi}\,(1+2x^2)}+\mathrm{erf}(x)+\lambda$
2	$-\frac{2e^{-x^2}x\,(15+20x^2+4x^4)}{5\sqrt{\pi}\,(3+12x^2+4x^4)}+\mathrm{erf}(x)+\lambda$
3	$-\frac{2e^{-x^2}x\,(105+210x^2+84x^4+8x^6)}{7\sqrt{\pi}\,(15+90x^2+60x^4+8x^6)}+\mathrm{erf}(x)+\lambda$

Table 8: Expressions for $I_m(x)$ in the Half Line Oscillator with $\alpha = -\frac{1}{2}$, for $m = 0, 1, 2, 3$ with $\omega_x = 2$.

Potentials and Ground State Wavefunction for Iso-Spectral Potential

m	$\hat{V}_m(x;\lambda,\alpha=\frac{1}{2})$
0	$x^2+\dfrac{2}{x^2}-8+\dfrac{128x^8}{\left(-3\sqrt{\pi}e^{x^2}(\mathrm{erf}(x)+\lambda)+4x^3+6x\right)^2}-\dfrac{32(x^2-2)x^3}{-3\sqrt{\pi}e^{x^2}(\mathrm{erf}(x)+\lambda)+4x^3+6x}$
1	$\Big[20\sqrt{\pi}e^{x^2}\left(24x^{10}+140x^8+430x^6+531x^4+480x^2-90\right)x(\mathrm{erf}(x)+\lambda)+25\pi e^{2x^2}\left(4x^8-36x^6-111x^4-108x^2+18\right)(\mathrm{erf}(x)+\lambda)^2+4\left(8\left(2x^6+4x^4-51x^2-255\right)x^6-2335x^4-2100x^2+450\right)x^2\Big]\div\left[x^2\left(-5\sqrt{\pi}e^{x^2}\left(2x^2+3\right)(\mathrm{erf}(x)+\lambda)+8\left(x^2+5\right)x^3+30x\right)^2\right]$
2	$\Big[28\sqrt{\pi}e^{x^2}\left(2\left(48x^8+776x^6+5208x^4+17780x^2+33235\right)x^6+55545x^4+23100x^2-3150\right)x(\mathrm{erf}(x)+\lambda)+49\pi e^{2x^2}\left(8\left(2x^6-12x^4-235x^2-885\right)x^6-8175x^4-3600x^2+450\right)(\mathrm{erf}(x)+\lambda)^2+4\left(4\left(4\left(4x^6+36x^4-237x^2-3962\right)x^4-66717x^2-122395\right)x^6-367255x^4-147000x^2+22050\right)x^2\Big]\div\left[x^2\left(2x\left(8x^6+84x^4+210x^2+105\right)-7\sqrt{\pi}e^{x^2}\left(4\left(x^2+5\right)x^2+15\right)(\mathrm{erf}(x)+\lambda)\right)^2\right]$

Table 9: Isospectral partner potentials $\hat{V}_m(x;\lambda,\alpha=\frac{1}{2})$ for $m=0,1,2$.

m	$\hat{V}_m(x;\lambda,\alpha=-\frac{1}{2})$
0	$\dfrac{12\sqrt{\pi}e^{x^2}x^3(\operatorname{erf}(x)+\lambda)+\pi e^{2x^2}\left(x^2-4\right)(\operatorname{erf}(x)+\lambda)^2+4\left(x^2+4\right)x^2}{\left(\sqrt{\pi}e^{x^2}(\operatorname{erf}(x)+\lambda)-2x\right)^2}$
1	$\left[12\sqrt{\pi}e^{x^2}x\left(6x^4+19x^2+4\right)\left(2x^2+3\right)(\operatorname{erf}(x)+\lambda)+9\pi e^{2x^2}\left(4x^6-28x^4-15x^2-16\right)(\operatorname{erf}(x)+\lambda)^2+4x^2\left(x^2+8\right)\left(2x^2+3\right)^2\right]\div\left[-3\sqrt{\pi}e^{x^2}\left(2x^2+1\right)(\operatorname{erf}(x)+\lambda)+4x^3+6x\right]^2$
2	$\left[20\sqrt{\pi}e^{x^2}x\left(12x^6+100x^4+153x^2+24\right)\left(4\left(x^2+5\right)x^2+15\right)(\operatorname{erf}(x)+\lambda)+25\pi e^{2x^2}\left(16x^{10}-96x^8-856x^6-1752x^4-567x^2-252\right)(\operatorname{erf}(x)+\lambda)^2+4x^2\left(x^2+12\right)\left(4\left(x^2+5\right)x^2+15\right)^2\right]\div\left[-20\sqrt{\pi}e^{x^2}x\left(4\left(x^2+3\right)x^2+3\right)\left(4\left(x^2+5\right)x^2+15\right)(\operatorname{erf}(x)+\lambda)+25\pi e^{2x^2}\left(4\left(x^2+3\right)x^2+3\right)^2(\operatorname{erf}(x)+\lambda)^2+4x^2\left(4\left(x^2+5\right)x^2+15\right)^2\right]$

Table 10: Isospectral partner potentials $\hat{V}_m(x;\lambda,\alpha=-\frac{1}{2})$ for $m=0,1,2$.

m	$\hat{\Psi}_m(x;\lambda,\alpha=\frac{1}{2})$
0	$\dfrac{2\sqrt{\frac{2}{3}}\sqrt{\lambda(\lambda+1)}e^{-\frac{x^2}{2}}x^2}{\sqrt[4]{\pi}\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x\left(2x^2+3\right)}{3\sqrt{\pi}}\right)}$
1	$\dfrac{2\sqrt{\frac{2}{5}}\sqrt{\lambda(\lambda+1)}e^{-\frac{x^2}{2}}x^2\left(x^2+\frac{5}{2}\right)}{\sqrt[4]{\pi}\left(x^2+\frac{3}{2}\right)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x\left(4x^4+20x^2+15\right)}{5\sqrt{\pi}\left(2x^2+3\right)}\right)}$
2	$\dfrac{2\sqrt{\frac{2}{7}}\sqrt{\lambda(\lambda+1)}e^{-\frac{x^2}{2}}x^2\left(4x^4+28x^2+35\right)}{\sqrt[4]{\pi}(4x^4+20x^2+15)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x\left(8x^6+84x^4+210x^2+105\right)}{7\sqrt{\pi}\left(4x^4+20x^2+15\right)}\right)}$
3	$\dfrac{2\sqrt{2}\sqrt{\lambda(\lambda+1)}e^{-\frac{x^2}{2}}x^2\left(8x^6+108x^4+378x^2+315\right)}{3\sqrt[4]{\pi}(8x^6+84x^4+210x^2+105)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x\left(16x^8+288x^6+1512x^4+2520x^2+945\right)}{9\sqrt{\pi}\left(8x^6+84x^4+210x^2+105\right)}\right)}$

Table 11: Isospectral partner potentials $\hat{\Psi}_m(x;\lambda,\alpha=\frac{1}{2})$ for $m=0,1,2$.

m	$\hat{\Psi}_m(x;\lambda,\alpha=-\frac{1}{2})$
0	$\dfrac{2\sqrt{\lambda(\lambda+1)}\,e^{-\frac{x^2}{2}}\,x}{\sqrt[4]{\pi}\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x}{\sqrt{\pi}}\right)}$
1	$\dfrac{2\sqrt{\lambda(\lambda+1)}\,e^{-\frac{x^2}{2}}\,x\left(x^2+\frac{3}{2}\right)}{\sqrt{3}\sqrt[4]{\pi}\,\left(x^2+\frac{1}{2}\right)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x\left(2x^2+3\right)}{3\sqrt{\pi}(2x^2+1)}\right)}$
2	$\dfrac{2\sqrt{\lambda(\lambda+1)}\,e^{-\frac{x^2}{2}}\,x\,(4x^4+20x^2+15)}{\sqrt{5}\sqrt[4]{\pi}\,(4x^4+12x^2+3)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x(4x^4+20x^2+15)}{5\sqrt{\pi}\,(4x^4+12x^2+3)}\right)}$
3	$\dfrac{2\sqrt{\lambda(\lambda+1)}\,e^{-\frac{x^2}{2}}\,x\,(8x^6+84x^4+210x^2+105)}{\sqrt{7}\sqrt[4]{\pi}\,(8x^6+60x^4+90x^2+15)\left(\operatorname{erf}(x)+\lambda-\frac{2e^{-x^2}x(8x^6+84x^4+210x^2+105)}{7\sqrt{\pi}\,(8x^6+60x^4+90x^2+15)}\right)}$

Table 12: Isospectral partner wavefunctions $\hat{\Psi}_m(x;\lambda,\alpha=-\frac{1}{2})$ for $m=0,1,2,3$.

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Uncertainty for Isospectral-Half Line Oscillator

λ	$m=0$	$m=1$	$m=2$	$m=3$
10^{-10}	0.687219	0.687258	0.687293	0.687325
10^{-5}	0.699465	0.702989	0.706010	0.708700
1	0.558179	0.572736	0.581874	0.588354
10^{5}	0.525237	0.534160	0.539833	0.543865
10^{10}	0.525237	0.534160	0.539833	0.543865
10^{15}	0.525237	0.534160	0.539833	0.543865

Table 13: Uncertainty values for $\alpha = \frac{1}{2}$ across different λ and m.

λ	$m=0$	$m=1$	$m=2$	$m=3$
10^{-10}	1.24144	1.24202	1.24239	1.24268
10^{-5}	1.22967	1.25449	1.27015	1.28271
1	0.651304	0.787014	0.869768	0.932633
10^{5}	0.583217	0.683772	0.747387	0.796065
10^{10}	0.583216	0.683770	0.747385	0.796063
10^{15}	0.583216	0.683770	0.747385	0.796063

Table 14: Uncertainty values for $\alpha = -\frac{1}{2}$ across different λ and m.